



Innovative Ecopure Technologies Private Limited



Performance Of High-Pressure RO
Membrane Technology in Industrial
Effluent Treatment Process to
Achieve Zero Liquid Discharge
(ZLD)

Get in touch...

Website : i-ecopure.com

E-mail : info@i-ecopure.com

Phone : +91 96006-02230

Address

No. 20/10, SF No. 182/1, 3, Elementary School
1st Street, Karuvampalayam,
Tiruppur – 641 604, Tamilnadu, India



Welcome to Innovative Ecopure Technologies Private Limited

About Us...

Innovative Ecopure Technologies (P) Ltd is a trusted leader in advanced wastewater and effluent treatment solutions, headquartered in Tirupur, India.

We specialize in turnkey design, engineering, construction, and operation & maintenance of treatment systems for industrial, municipal, and residential applications.

Our technology-driven approach enables clients to achieve regulatory compliance, operational efficiency, and sustainability goals.

Our Strength...

We are a key player in the waste-water treatment, managing eight top-performing CETPs, ranked by the Indian Institute of Technology – Chennai (IIT - Chennai). With the capability to treat over 70 MLD of effluent, we demonstrate strong expertise in handling large-volume wastewater treatment systems.

All CETP operations are fully transparent and undergo regular online monitoring by the Tamil Nadu Pollution Control Board, reinforcing our credibility and demonstrating our proven expertise in the field.

Our team comprises highly skilled professionals with extensive experience in the water and wastewater treatment industry, further strengthening our ability to operate and manage high-capacity CETPs with efficiency and reliability.

Our dedicated R&D team proactively addresses issues at the earliest stages, preventing operational disruptions and avoiding unnecessary expenditure.

Pioneering High-Pressure RO Technology...

Innovative Ecopure Technologies Pvt. Ltd. was the first to implement high-pressure RO systems for textile effluent recycling across Tamil Nadu. These systems operate with high-pressure and booster pumps that elevate feed pressure to 1300–1500 psi, enabling high recovery and efficient water reuse.



About the Textiles...

The textile industry is one of India's key income-generating and rapidly growing sectors. As a major consumer of chemicals and water, it uses an estimated 100–150 litres of water per kilogram of textile produced. This results in significant volumes of wastewater being generated. When discharged without proper treatment, textile effluents contribute to serious environmental challenges, particularly water pollution in surrounding ecosystems.

Textile industry effluent is a significant environmental concern due to its high load of organic compounds, elevated BOD and COD levels, dissolved salts, and diverse dyes. These pollutants reduce oxygen availability and block light penetration in water bodies, disrupting aquatic ecosystems.

Effective treatment is essential to remove contaminants and enable reuse of treated water in dyeing processes. Treatment typically involves preliminary, primary, secondary, and tertiary (advanced) systems. While the first three stages address BOD, COD, and color removal, reverse osmosis is crucial for achieving the quality required for wastewater recycling.



RO Membrane Technology

Reverse osmosis (RO) is a key technology in industrial wastewater treatment. Its semipermeable membrane contains microscopic pores that allow only water molecules to pass, effectively rejecting dissolved salts, particles, colloids, and microorganisms. RO produces two streams: permeate (treated water) and concentrate (reject). The ratio between these streams defines the system's recovery rate, a key performance indicator.

Effective pretreatment is essential to protect the membrane from fouling, scaling, and premature failure, ensuring consistent operational efficiency.

Industrial RO systems are typically categorized into low-pressure RO (operating at 400–600 psi) and high-pressure RO, which requires 1300–1500 psi to function. The primary distinction between them is the operating pressure needed to achieve the desired level of purification.

Low - Pressure RO Membrane Technology at Sirupooluvapatti CETP (Up to 2017)

Preliminary treatment involves screening to remove large suspended solids, followed by collection. Primary treatment is carried out through equalization with continuous aeration and subsequent neutralization. The neutralized effluent then enters secondary treatment, which removes BOD, COD, color, and turbidity.

Post-biological treatment, the effluent undergoes chlorine oxidation for further color removal, followed by pressure sand filtration (PSF) and microfiltration (MF). The pretreated water is then processed through RO stages 1, 2, and 3, with feed characteristics and recovery rates summarized in Table 1. Reject from these stages is fed to a fourth RO stage, achieving a combined recovery of up to 85%.

The final RO reject is softened to reduce hardness and then processed through MVRE and MEE, achieving recovery rates of 7.25% and 1.5%, respectively. Approximately 50% of the MVRE reject (about 4% of the total volume) is recovered as brine with 108,000–120,000 ppm TDS, which is supplied back to member units for reuse in dyeing operations.

Parameter	Low Pressure RO feed / Recovery (Stage1, 2, 3) (80% Recovery)	Low pressure RO Feed / Recovery (Stage 4th) (5% Recovery)	MVRE feed / Recovery (7.25%)	MVRE Reject - Brine Recovery (4%)	MVRE Reject - MEE Feed / Recovery (1.5%)	MEE Reject / Pusher Centrifuge Feed	Final Recovery
Flow Rate (KLD)	2400/1920	480/120	360/180	96	84/36	48	2352 (97.75)
Power	2KW/KL		20KW/KL		50KW/KL		
TSS (ppm)	<5	<25	<35	<70	<70	--	--
Turbidity (NTU)	0	<2	<3	<6	<6	--	--
BOD (ppm)	<80	<400	<550	<1100	<1100	--	--
COD (ppm)	<200	<1000	<1350	<2700	<2700	--	--
TDS (ppm)	8000-9000	40000-45000	54000 - 60000	108000 -120000	108000 -120000	300000-350000	200-300

High - Pressure RO Membrane Technology at Sirupooluvapatti CETP

High-pressure RO has been successfully implemented at SCETP since 2021 and continues to operate efficiently. The influent and effluent characteristics are presented in Table 2. The system achieves a significantly higher recovery rate of 94%, compared with 85% achieved by low-pressure RO.



Advantages

Compared to Mechanical Vapor Recompression (MVRE), high-pressure RO offers several key benefits:

- Higher water recovery rate
- Lower manpower requirements
- Reduced maintenance costs
- Consistent and stable operation

Running Cost

Operating Cost: Approximately Rs. 180 per KL, inclusive of power, chemicals, and manpower.

Parameter	Low Pressure RO feed / Recovery (Stage1, 2, 3) (82% Recovery)	High pressure RO Feed / Recovery (Stage 4, 5 & 6) (12% Recovery)	High pressure RO-stage 4, 5 & 6 Reject/MEE feed	Brine Recovery rate 3% (50% of High-Pressure RO Reject)	MEE Feed (50% of High-Pressure RO Reject) & Recovery (1.5%)	MEE Reject/Pusher Centrifuge Feed	Final Recovery
Flow Rate (KLD)	4000/3280	720/480	240	120	120/60	60	3940 (98.5%)
Power	2KW/KL	8KW/KL	--	--	50KW/KL	--	--
TSS (ppm)	<5	<5	<20	<20	<20	--	--
Turbidity (NTU)	0	0	<2	<2	<2	--	--
BOD (ppm)	<80	<450	1500	1500	1500	--	--
COD (ppm)	<200	<1100	3700	3700	3700	--	--
TDS (ppm)	8000-9000	45000 - 50000	150000-165000	150000-165000	150000-165000	300000-350000	200-300

Schematic flow:

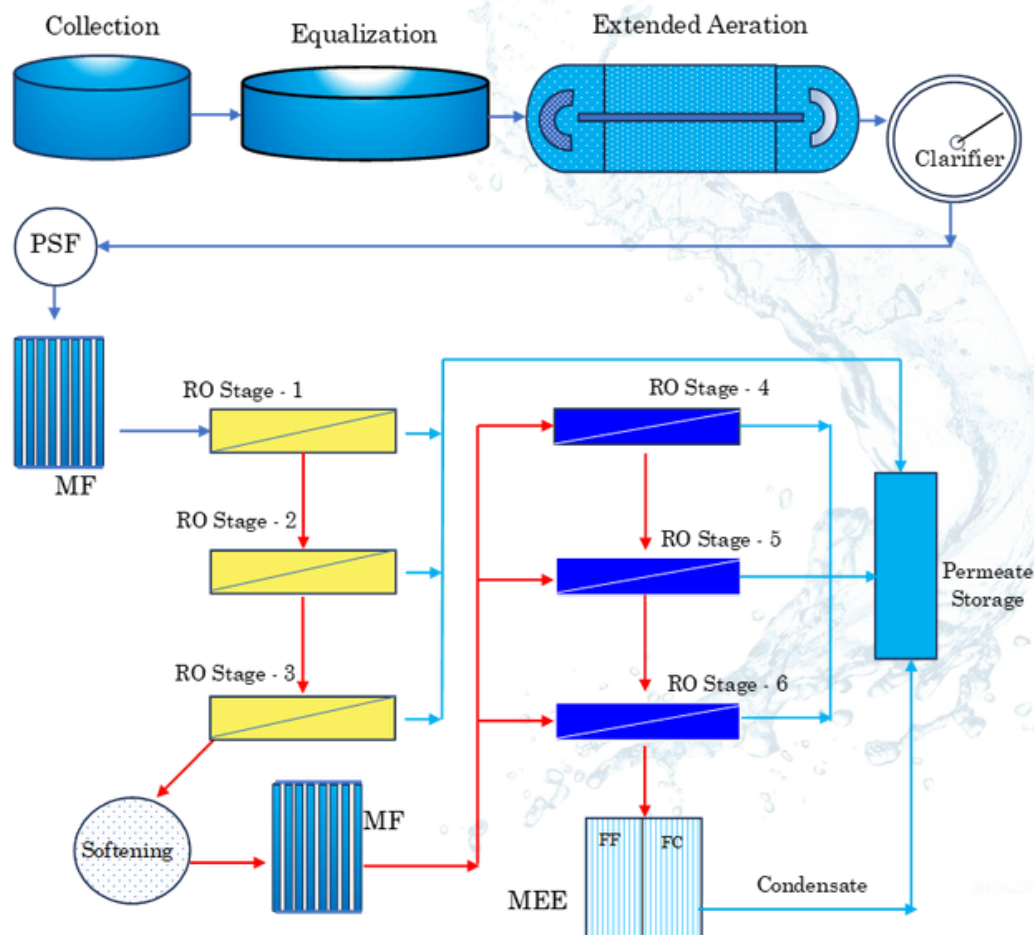
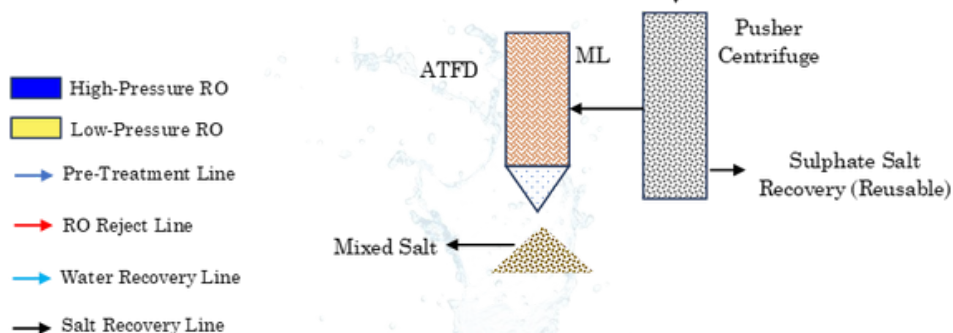


Fig. 3. Process flow of High-Pressure RO



Conclusion:

The findings of this case study indicate that high-pressure reverse osmosis (RO) technology offers clear advantages over low-pressure RO combined with MVRE. High-pressure RO delivers a significantly higher recovery rate while operating at a substantially lower overall cost, making it a more efficient and economical solution for textile effluent treatment and recycling.

Your challenges, Our expertise - The Perfect Solution